

PUBERTY AND GONADAL HORMONES

DR HUMAIRA FAZAL

LECTURER

DIKIOS, DUHS

LEARNING OBJECTIVES:

- Define puberty and describe its major stages.
- Explain the hormonal regulation of puberty and the role of the hypothalamic–pituitary–gonadal axis.
- Describe the functions of GnRH, LH, and FSH during puberty.
- Explain the production and actions of testosterone, estrogen, and progesterone.
- Describe the physical and reproductive changes occurring during male and female puberty.
- Explain the role of gonadal hormones in secondary sexual characteristics.

PUBERTY:

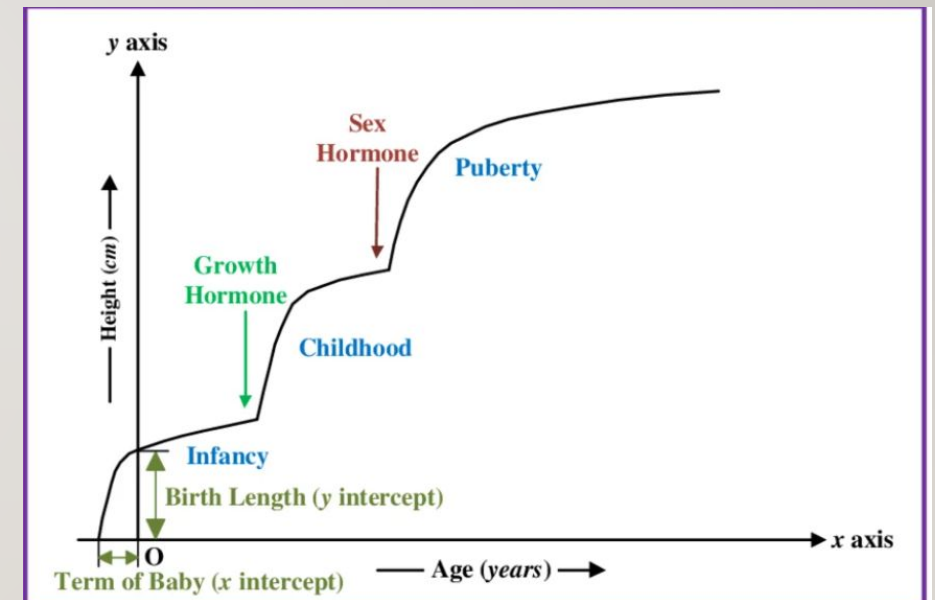
- Puberty means the onset of adult sexual life
- The period of puberty is caused by a gradual increase in gonadotropic hormone secretion by the pituitary, beginning in about the eighth year of life
- GIRLS: Average age of onset of puberty is 10.5 years(normal range from 8-13 years.
- BOYS:Average onset of puberty is 11 years(normal range from 9-14 years)

-
- Puberty starts when a part of your child's brain called the hypothalamus begins producing a hormone called gonadotropin-releasing hormone (GnRH).
 - Their hypothalamus sends GnRH to another part of their brain called the pituitary gland. GnRH stimulates your child's pituitary gland to release two more hormones — luteinizing hormone (LH) and follicle-stimulating hormone (FSH).
 - These hormones travel to the sex organs (gonads) — the ovaries or testes. This triggers your child's gonads to begin releasing sex hormones — either estrogen or testosterone. These messenger hormones cause the telltale signs of puberty to begin.

PHYSIOLOGY OF PUBERTY:

PERIOD OF INFANCY AND CHILDHOOD:

- The hypothalamic-pituitary system (“the gonadostat”) is highly sensitive to negative feedback of estrogen.
- (estradiol concentrations are low as 10pg/ml)
- Gonadotropins are gonadal steroids remain at very low levels until age 6-8
- FSH and LH remain low but are still secreted in pulsatility.



PUBERTY AND REGULATION OF ITS ONSET:

PREPUBERTAL PERIOD:

3 critical changes in the endocrine state of childhood emerge

- 1) Adrenarche
- 2) Decreasing repression of the “gonadostat”
- 3) Gradual amplification of the GnRh- gonadotropin interaction

LATE PREPUBERTAL PERIOD:

- **Adrenarche (Pubarche):** Occurs during the late prepubertal period (around ages 8–11). Increasing in production of adrenal androgens resulting in growth of pubic hair, axillary hair, acne and body odor. Precedes menarche by 2 years.
- **Initiating mechanism:** suggested factors- pituitary adrenalandrogen stimulating factor
- **Activation of the HPG Axis:** The onset of true puberty is signaled by the pulsatile release of Gonadotropin-Releasing Hormone (GnRH) from the hypothalamus. This stimulates the pituitary gland to release LH and FSH, driving sex steroid production (estrogen in females, testosterone in males).
- **Physical & Secondary Sexual Changes:**
 - **In Girls:** Typically begins with *thelarche* (breast bud development around age 10.5), followed by a peak growth spurt, and eventually *menarche* (first menstrual bleed) 2–3 years later.
 - **In Boys:** Typically begins around age 11.5–12 with testicular enlargement, followed by progressive secondary sexual characteristics.

STAGES OF PUBERTY:

Puberty: Girls

- **Thelarche:** (Breast budding) usually first sign.
It often begins unilaterally (on one side first) before progressing bilaterally.
- **Menarche:** usually 2-3 years after breast development
- **Growth spurt:** The peak velocity of the female growth spurt occurs *before* menarche.
- **Pubic and axillary hair growth:** sign of adrenal androgen secretion
- Starts at similar stage of apocrine gland sweat production and associated with adult body odor
- Estrogen do not greatly affect hair distribution .However, hair does develop in the pubic region and in the axillae after puberty.
- Androgens formed in increased quantity of the female adrenal gland after puberty area responsible for this

FEMALE PUBERTY

Puberty is a time of change and development — you are growing up! Puberty begins around 9–14 years of age. Changes happen in the body caused by chemicals called hormones. It's important to remember everyone is different and not to compare your journey to adulthood with others. However, there are common changes that happen during puberty.

BODY SIZE

Arms, legs, feet, and hands grow at a faster rate than other body parts.



EMOTIONS

Fear, frustration, anger, sadness, mood swings and so much more are all feelings you may encounter during puberty. It is a great change physically and emotionally.



BREASTS

Breasts grow larger and develop. Sometimes one is larger than the other — that's natural and will balance out. They develop and a bra may be needed for support.



SWEATING

Body odour can develop, so you may need to shower more regularly.



BODY SHAPE

Breasts will continue to grow, hips will widen and the waists get smaller.



SKIN

Skin sometimes becomes oily and spots and/or acne may appear.



HAIR

Hair will start to grow on the pubic regions, underarms and legs.



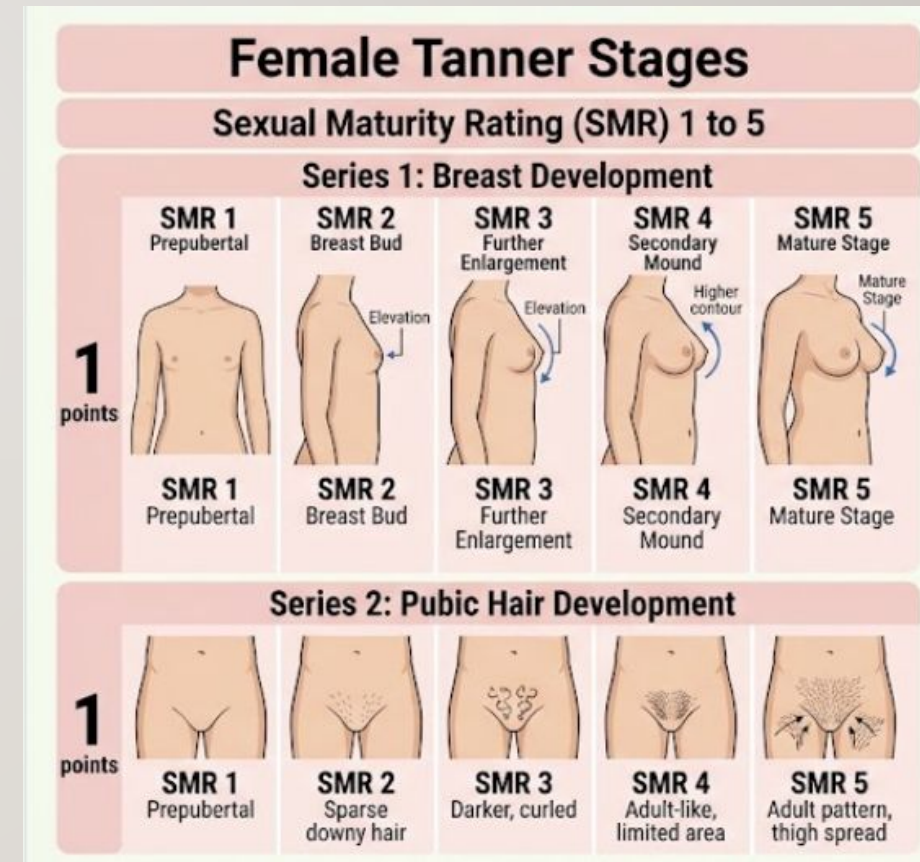
MENSTRUATION

Menstruations usually starts between the ages of 9–14. A layer of blood and tissue builds up in the uterus and is discharged monthly. The flow varies from person to person.



PUBERTAL STAGES (TANNER): FEMALE

- **P1 (Prepubertal):** Represents the prepubertal state with no breast tissue development or pubic hair.
- **P2:** Marked by early development of a subareolar breast bud, with or without small amounts of pubic and axillary hair.
- **P3:** Characterized by an increase in the size of palpable breast tissue and areolae, along with increased dark, curled pubic and axillary hair.
- **P4:** Features breast tissue and areolae elevated to form a secondary mound above the breast contour, alongside adult-type pubic hair that has not yet spread to the medial thighs.
- **P5:** Represents mature adult breasts with projection of the papilla only, and pubic hair extending to the upper inner thighs.



MENARCHE;

- Age of menarche chiefly controlled by genetics
- Environmental factor important: improving nutrition , produces taller, heavier girls
- During puberty oestradiol levels fluctuate widely (reflecting successive waves of follicular development that fail to reach ovulatory stage)
- Menses are usually anovulatory for 12-18 months after menarche because the LH surge is not potent enough at these time to cause ovulation.

MENSTRUAL CYCLE :

Changes in the Body During the Menstrual Cycle

Menstrual Phase Days 1-5

Progesterone levels quickly withdraw, triggering the uterine lining to shed and the period to begin.

Luteal Phase Days 17-28

Raised hormone levels help thicken the endometrium. However, if there isn't a fertilized egg, the endometrium prepares to shed.



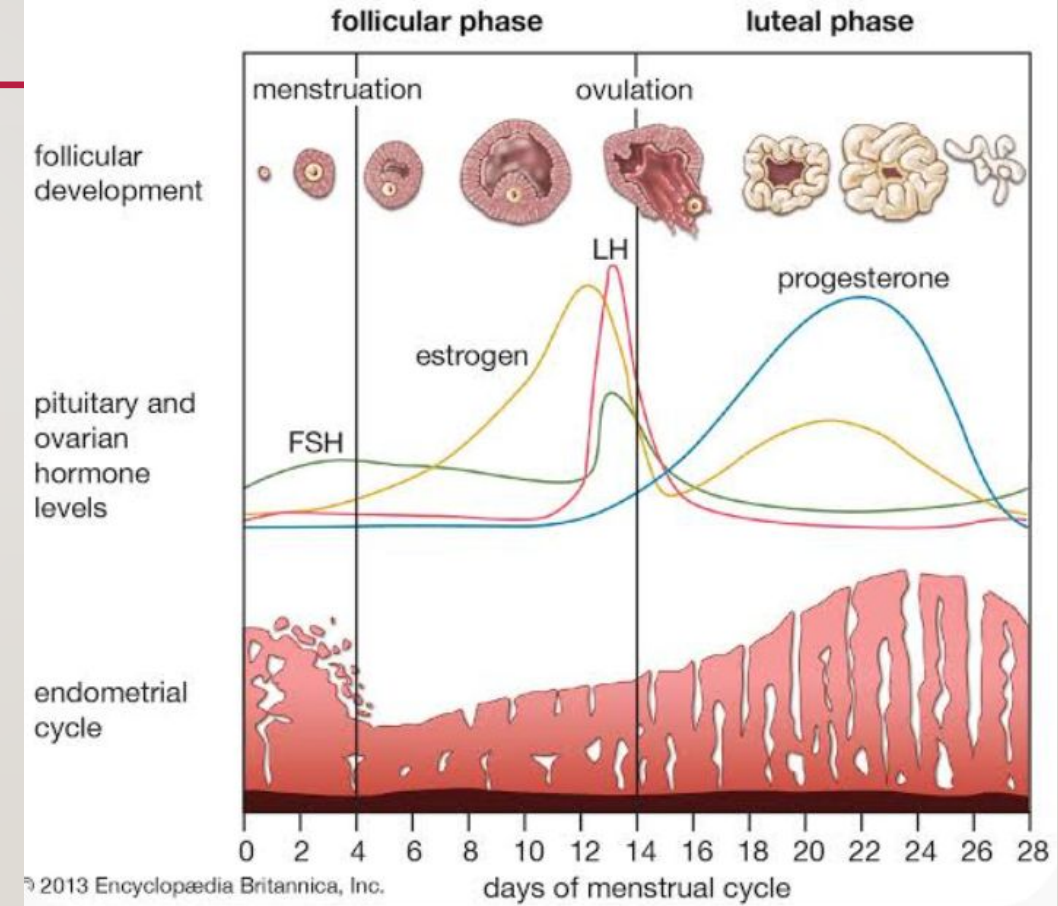
Follicular Phase Days 1-14

Rising levels of follicular stimulating hormone cause several ovarian follicles to undergo maturation.

Ovulation Days 12-16

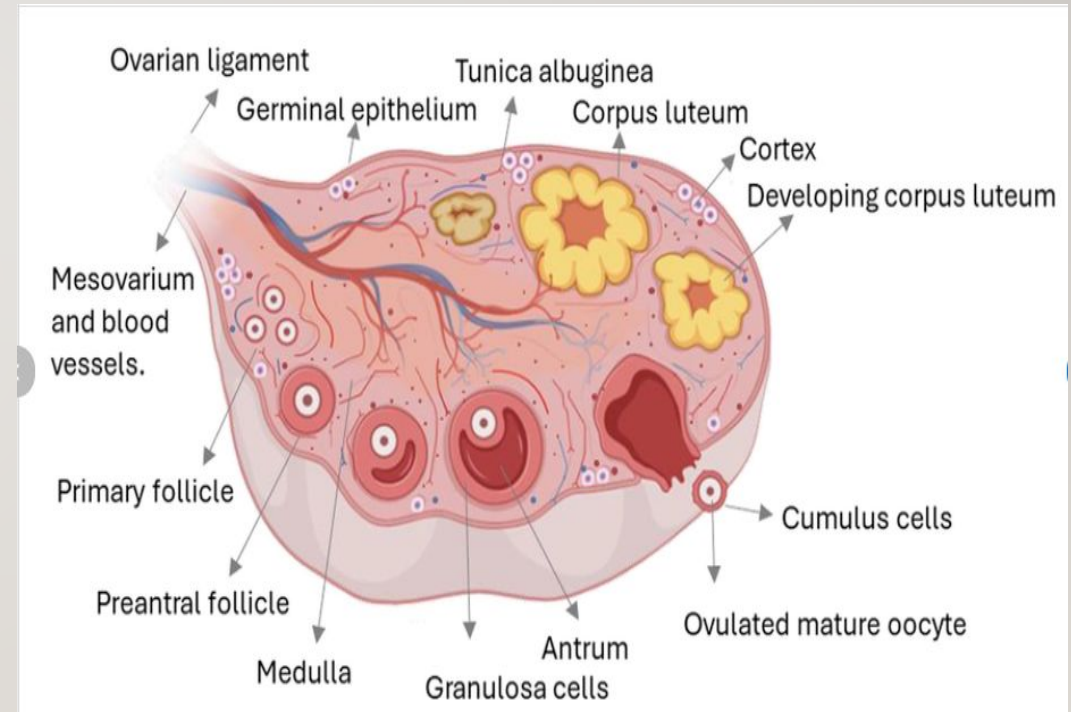
Luteinizing hormone surges causing a fully-matured egg to be released into the fallopian tube.

The menstrual cycle



OVARIAN DEVELOPMENT:

- Rising level of plasma gonadotropins stimulate ovary to produce increasing amounts of oestradiol –secondary sex characteristics
- Prepubertal volume-0.3-0.9cm³
- >1.0cm³ indicates puberty has begun
- During puberty- rapid increase in size
- Mean post pubertal volume 4cm³



Development of Uterus

- Prepubertal uterus is tear-drop shaped
- Neck and isthmus account for up to 66% of uterine volume
- Following production of oestrogens – uterus becomes pear shaped
- Uterine body increases in length (max 5 – 8cm) and thickness (proportionately more than cervix)

PUBERTY: BOYS

- First sign often go unnoticed
- Testicular enlargement(12-13 years)
- Prepubertal testis- 2mls diameter
- Puberty begins when volume reaches 4mls
- Penile and scrotal enlargement occur approximately 1 year after testicular enlargement
- Pubic hair appear at same time
- Pubertal Growth Spurt: occur later in female but greater

Testosterone required in large concentration to produce same anabolic effect

MALE PUBERTY

The changes that happen in the body are caused by chemicals called hormones. Puberty for boys usually starts between ages 10–16. It is a time of great change and can be a confusing and emotional time. Puberty is the beginning of your journey from a boy into a man.

While it can be challenging, it is also one of the most natural things in the world. It is important to keep talking about how you feel to a trusted adult, like a parent, guardian or another person.

BODY SIZE

Arms, legs, feet, and hands grow at a faster rate than other body parts.



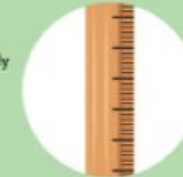
BODY SHAPE

Arms, legs and feet will grow faster than the rest of your body. Your shoulders broaden and muscles and weight are gained.



GROWTH SPURT

You will experience a growth spurt in puberty, which typically ends around the age of 16.



VOICE

Your voice will change and become deeper. This usually occurs just after a big growth spurt.



SKIN

Your skin may become oily, and you will sweat more, meaning a development of acne or spots on your back and/or face.



EMOTIONS

Puberty will affect your mental health. Feelings of fear, anger, excitement and embarrassment become more common.



HAIR

Hair begins to grow on your underarms, above the penis, arms, face and sometimes the chest.



ERECTIONS

Erections are completely normal and happen due to increased hormones.



PENIS

Your penis will grow longer and thicker and the testicles will grow larger too.



SPERM

Sperm begin to be produced by your body. During an erection, you may experience ejaculation which is natural and normal.



PUBERTAL STAGE (TANNER) MALE :

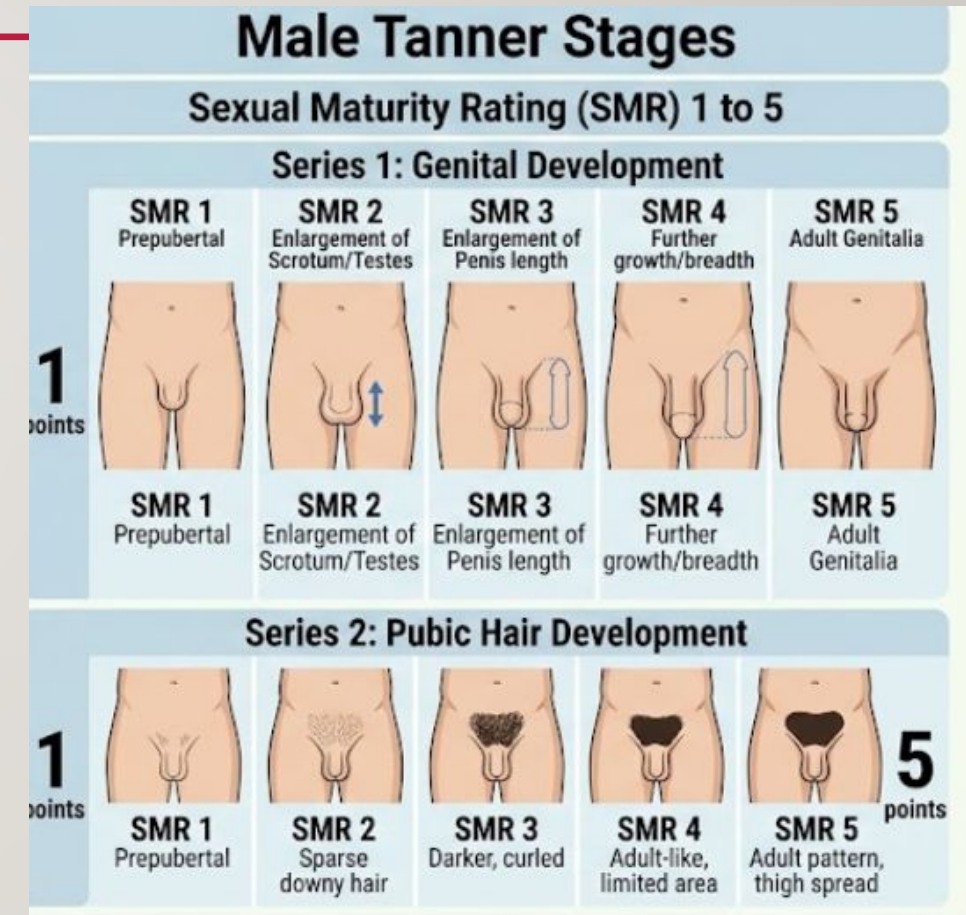
P1: Prepubertal, testicular volume < 2mls

P2: Enlargement of scrotum and testes. Scrotum slightly pigmented. Few long dark pubic hair

P3: lengthening of penis. Further growth of testes and scrotum. Pubic hair darker, coarse, more curled

P4: Penis increases in length and breadth, increased pigmentation of scrotum. Adult type pubic hair, no spread to medial thigh

P5: Genitalia adult in size and shape, pubic hair spread to inner thighs



Abnormal Puberty

Delayed Puberty: Defined as the absence of pubertal onset by <8years in females or < 9 years in males.

- **Hypogonadotropic Causes:** Constitutional delay (familial/sporadic), chronic illness (CF, Crohn's, renal failure), malnutrition (anorexia, coeliac disease), excessive exercise, pituitary/hypothalamic tumors (craniopharyngioma), syndromes (Prader-Willi, Laurence-Moon-Biedl), and hypothyroidism.
- **Hypergonadotropic Causes:** Congenital (Turner Syndrome, Klinefelter Syndrome) or acquired (chemotherapy, radiation, surgery, testicular torsion, trauma, infection, autoimmunity).
- Maybe associated with growth spurt

ABNORMAL PUBERTY

Precocious Puberty

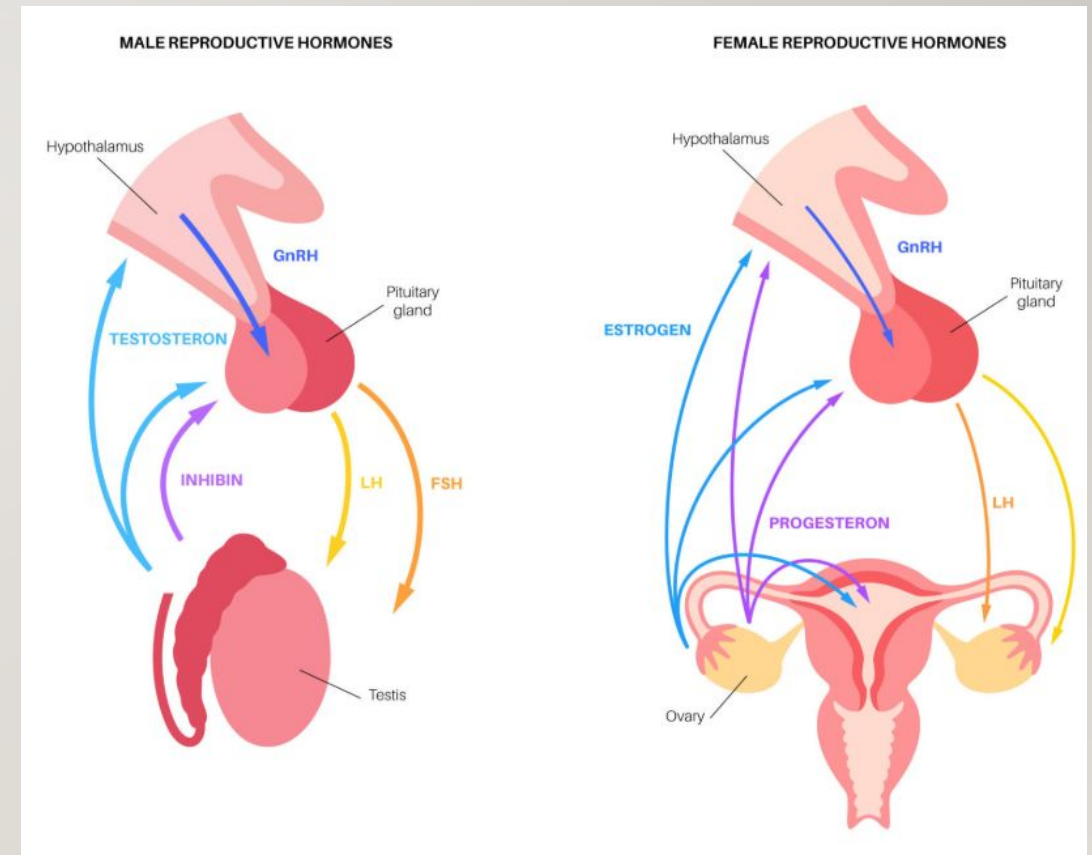
- Defined as the development of secondary sexual characteristics before age 8 in females and age 9 in males.
- 5 times more common in females, where it is usually a benign central process.
- Less common in males, but approximately 50% of male cases are due to an underlying pathological cause

Benign Variants

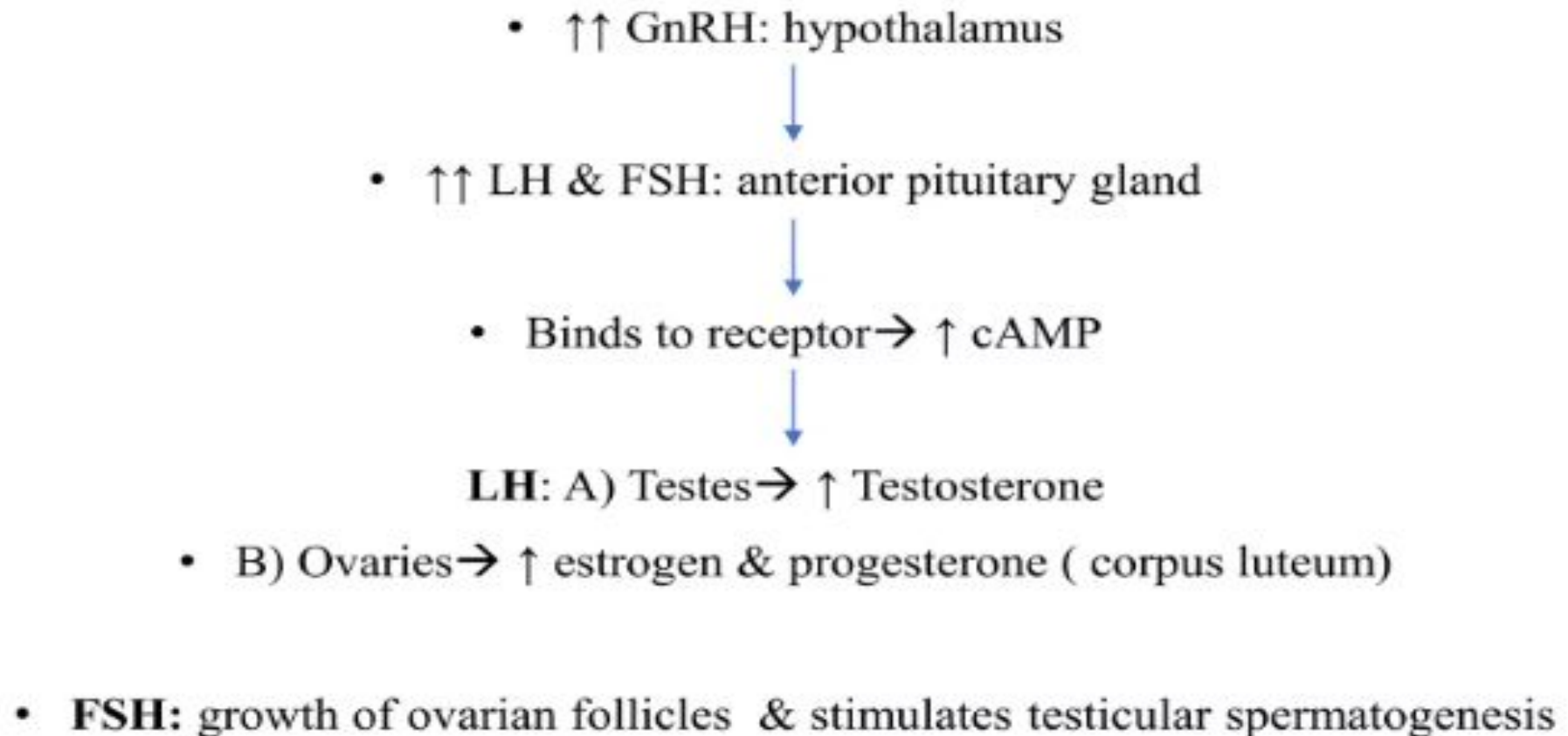
- **Premature Thelarche:** Isolated breast development in girls < 3 years} with spontaneous regression.
- **Premature Pubarche:** Isolated pubic hair development in children < 7 years} due to mid-childhood adrenal androgen secretion.

GONADAL HORMONES

- **Gonads:** are the primary reproductive organs which produce the gametes (egg or ovum); a pair of testes (singular = testis) produces sperms in males and a pair of ovaries produces ovum in females.
- Gonadal hormones are hormones produced by the gonads, and include both steroid and peptide hormones. The major steroid hormones include **ESTROGEN** and **PROGESTERONE** from the **OVARY**, and **TESTOSTERONE** from the **TESTIS**. The major peptide hormones include **ACTIVINS** and **INHIBINS**.



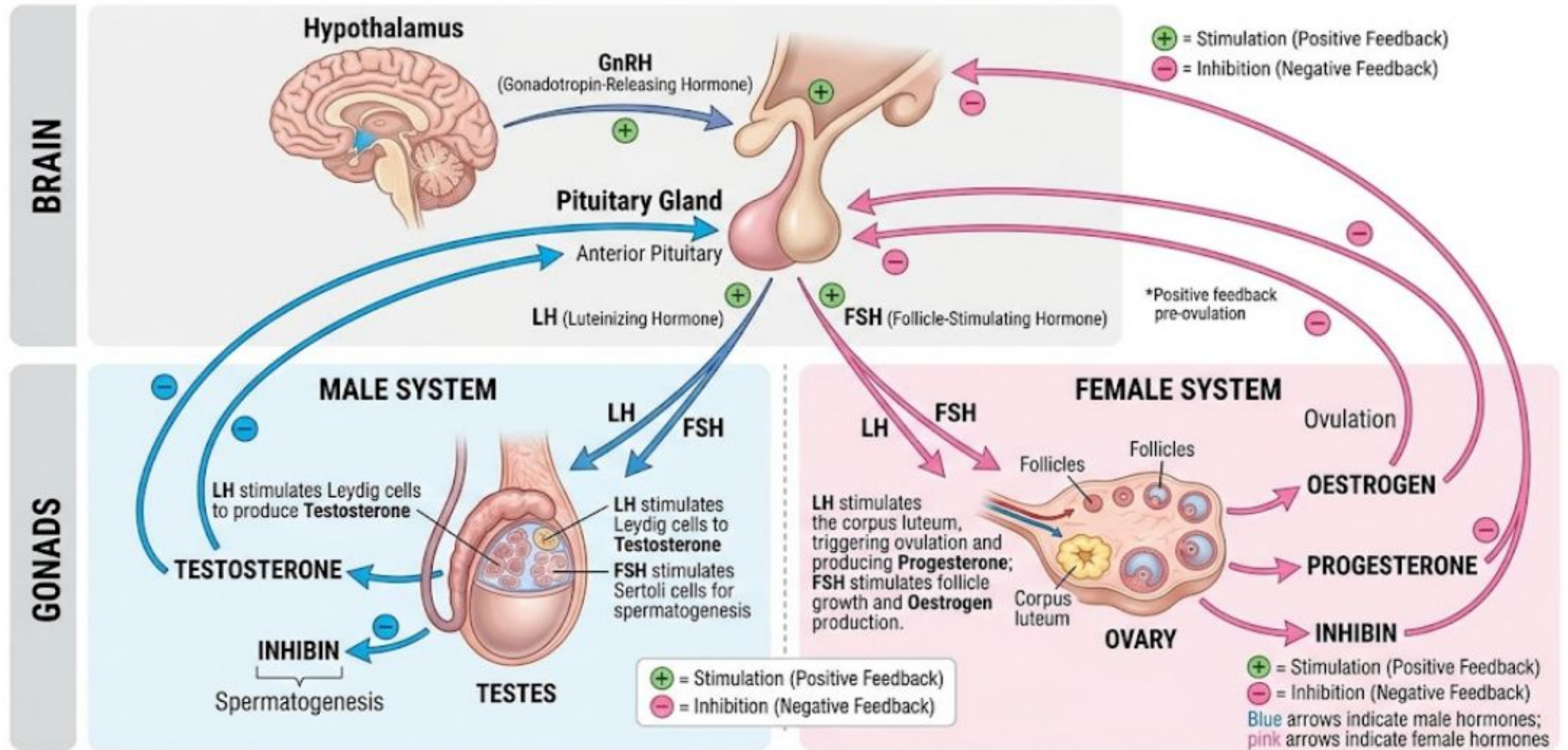
- From: testes & ovaries
- Aim: sexual differentiation & reproduction.



HYPOTHALAMIC-PITUITARY-GONADAL (HPG) AXIS

- **Reactivation of the HPG Axis** The hypothalamic-pituitary-gonadal (HPG) axis—which is well-developed at birth but dormant during early childhood—reactivates at the onset of puberty.
- **Pulsatile GnRH Release** Mature GnRH neurons release GnRH in a pulsatile manner into the hypophyseal portal system, which transports it to pituitary gonadotropes.
- **Secretion of Gonadotropins (FSH & LH)** In response to GnRH pulses, the pituitary synthesizes and releases FSH and LH into systemic circulation in a matching pulsatile pattern.
- **Gonadal Maturation & Gamete Production** Circulating gonadotropins stimulate full gonadal development, sex steroid hormone secretion, and gamete production in both sexes.
- **Initiation of Puberty & Fertility** This cascade successfully initiates puberty and establishes ongoing reproductive physiology and fertility

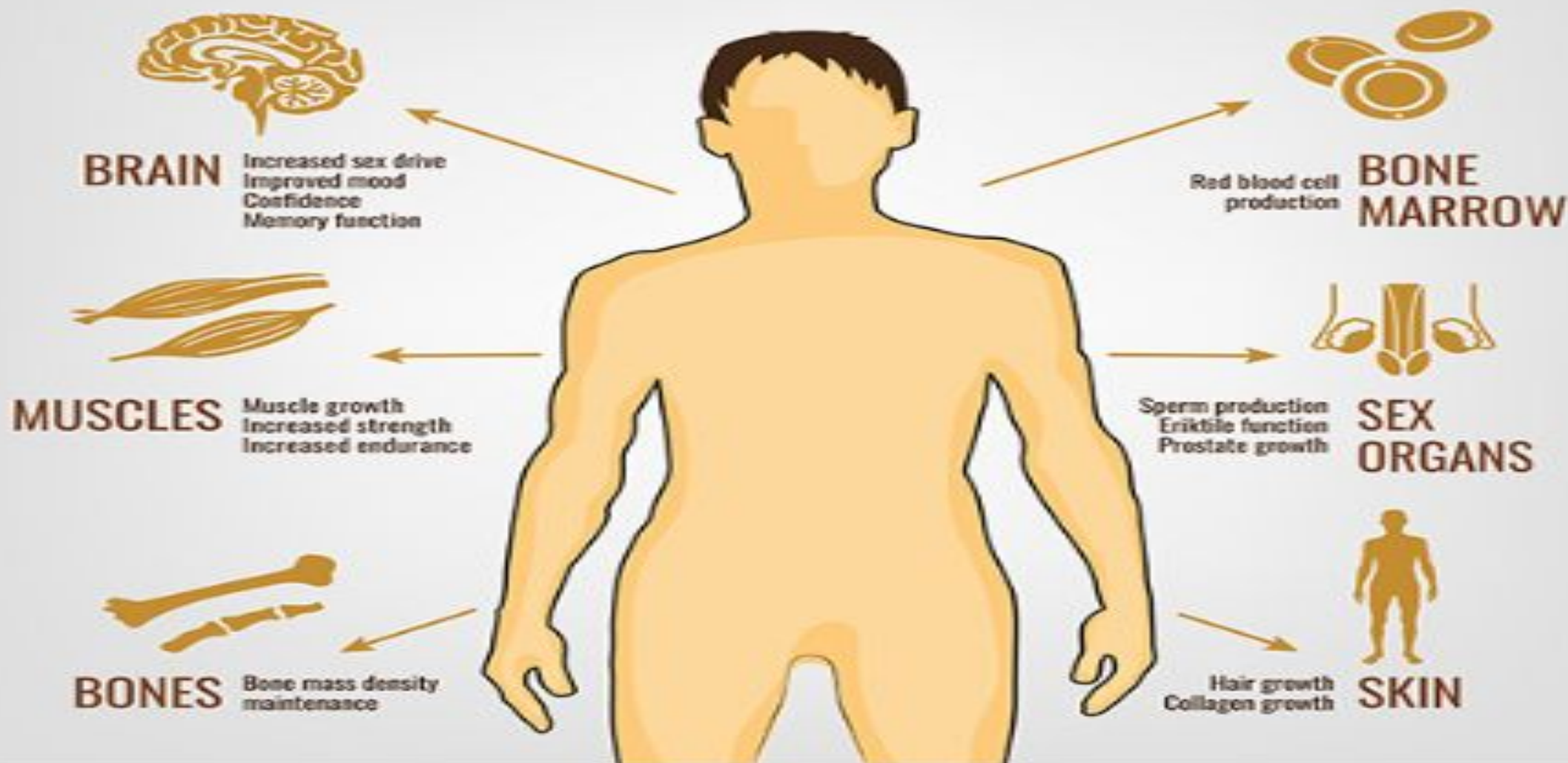
DIAGRAM OF THE HYPOTHALAMIC-PITUITARY-GONADAL (HPG) AXIS & HORMONE REGULATION



TESTOSTERONE IN PUBERTY & ADULTHOOD

- **REGULATION OF TESTOSTERONE SECRETION:** During fetal life, the testosterone secretion from testes is stimulated by **human chorionic gonadotropin**, which has the properties similar to those of luteinizing hormone.
- **Pubertal Effects:** Drives the transition to manhood by developing male sex organs, facial and body hair, deepening the voice, increasing height and muscle mass, and growing the Adam's apple.
- ***In Adults* Luteinizing hormone (LH) or interstitial cell stimulating hormone (ICSH)** stimulates the Leydig cells. **Adult Functions:** Maintains ongoing sperm production, muscle strength and mass, bone density, and blood cell formation.
- **Regulation:** The hypothalamus releases GnRH to stimulate the pituitary gland to release FSH and LH. LH triggers testosterone production, while a negative feedback loop alerts the pituitary to reduce LH if testosterone levels become too high.

THE INFLUENCE OF TESTOSTERONE



Ovarian Hormones

- The ovaries are the key source of gonadal hormones during a woman's reproductive years (puberty to menopause).
- The pituitary gland regulates these hormones via follicle-stimulating hormone (FSH) and luteinizing hormone (LH), driving the menstrual cycle.
- Each cycle, an ovarian follicle matures, boosts estrogen production, releases an egg, and becomes the progesterone-secreting corpus luteum.
- Without fertilization, the corpus luteum degenerates, triggering the shedding of the uterine lining (menstruation) and restarting the cycle.

ESTROGEN HORMONES:

Function: Promotes female sexual development and growth/maturation of primary and secondary sex characteristics. It helps regulate cholesterol levels. Influences the brain, including mood regulation, as well as the condition of bones, skin, and various other tissues

- **In Males:** Present in smaller amounts, assisting in sperm maturation.
- **Pubertal Changes:** Drives breast growth, pubic and underarm hair development, and the onset of menstrual cycles.
- **Reproductive Role:** Controls menstrual cycles and supports childbearing

The human body generates three primary forms of estrogen:

- Estradiol (E2): Predominantly found in women during their reproductive years.
 - Estriol (E3): The chief estrogen produced throughout pregnancy.
 - Estrone (E1): The main estrogen present after menopause, once menstrual cycles have ceased.
- 

PROGESTERONE

- Progesterone is a hormone classified as a steroid.
- This hormone is produced by the corpus luteum, a short-lived endocrine gland formed in the ovary after ovulation, during the latter half of the menstrual cycle.
- Its primary role is to prepare the uterine lining (endometrium) for a possible pregnancy by stimulating it to thicken, creating a supportive environment for a fertilized egg to implant.
- Additionally, progesterone suppresses uterine muscle contractions, preventing the expulsion of the implanted embryo.
- If fertilization does not occur, the corpus luteum degenerates, causing progesterone levels to drop, which then triggers the onset of menstruation.

-
- Progesterone sustains blood vessel growth in the endometrium to nourish the fetus if conception occurs.
 - The placenta secretes progesterone after developing, supporting the corpus luteum and keeping hormone levels high to prevent new egg production.
 - Progesterone prepares the breasts for milk production.

ACTIVINS AND INHIBIN

- **Activins and Inhibins:** Protein hormones made by gonads (testes and ovaries).
- **Role:** Regulate Follicle Stimulating Hormone (FSH) from the pituitary gland.
- **Activin:** Stimulates the pituitary gland to release more FSH.
- **Inhibin:** Inhibits FSH production and acts as its antagonist.
- **Feedback:** Inhibin sends negative feedback to reduce FSH secretion from the pituitary.

ANDROGENS

Testosterone
Major male
sex hormone

Produced by
Leydig cells

ESTROGENS

Predominant
ovarian
hormone

Follicle and
corpus luteum
of ovary

PROGESTERONE

Produced
during luteal
phase of
menstrual
cycle and also
during
pregnancy